

Juan David Muñoz-Bolaños

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Mühlbauer GmbH & Co. KG

Frau Tugce Zabun

Josef-Mühlbauer-Platz 1

93426 Roding

Deutschland

Subject: Application for Application Engineer Image Processing (m/f/d)

Dear Ms. Zabun,

As a physics engineer specialized in high-precision metrology and industrial machine vision, I am applying for the Application Engineer position at your Roding site. Your focus on hardware and optical development over pure software programming perfectly aligns with my passion for building and optimizing physical optical systems.

In my previous industrial experience at INTECOL S.A.S., I successfully integrated vision systems using smart cameras and advanced illumination methods for a car subdivision of Renault, where I was responsible for the hardware-software integration and custom illumination. During my PhD, I pushed this further by building automated wavefront sensing setups from scratch, qualifying high-end optomechanical components from vendors like Thorlabs and Edmund Optics. This hands-on experience in precise system alignment and hardware validation enables me to bridge the gap between complex optical designs and reliable industrial assembly.

With a strong background in QA processes and a proven ability to design rigorous test structures, I would be glad to help optimize Mühlbauer's vision systems. I am fully integrated into the DACH work culture and, as requested, highly willing to support international service assignments. I look forward to contributing my technical expertise to your team.

Sincerely,

Juan David Muñoz-Bolaños

Application Engineer Image Processing

Medizinische Universität Innsbruck

Juan David Muñoz-Bolaños

Application Engineer Image Processing

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Professional Summary

Dedicated R&D Optical Engineer with a robust background in combining **High-Precision Optical Metrology** with **Industrial Machine Vision** systems. Expert in developing advanced illumination concepts, real-time image processing algorithms (HALCON/Python), and hardware-software integration for automated inspection units. Proven problem-solving skills with a track record of bridging the gap between complex optical research and reliable industrial deployment.

TECHNICAL COMPETENCIES

Machine Vision Hardware

Mastering of CMOS/CCD camera integration, lens selection (telecentric, macro), and custom illumination design (Dark-field, Bright-field, coaxial).

Image Processing Libraries

Advanced proficiency in Halcon and Cognex VisionPro for real-time inspection, part recognition, and measurement. Familiar with OpenCV and Python-based vision pipelines.

System Alignment & Justage

Hands-on experience in the precise alignment of complex opto-mechanical setups (Laser sources, scanners, detectors). Expert in qualifying components from market-leading vendors.

Automation & Integration

Skilled in automating optical systems with Python, C++, and LabVIEW. Experienced in ensuring Gage R&R and measurement reliability in harsh industrial environments.

R&D Methodology

Experience bringing systems from concept to series prototypes, including ZEMAX modeling, procurement management, and technical documentation.

PROFESSIONAL EXPERIENCE

INTECOL S.A.S.

Junior Machine Vision System Engineer

Medellín, Colombia

Mar 2018 – Mar 2019

- **Vision Pipeline Engineering:** Designed and implemented Halcon-based beverage bottle inspection systems; developed custom illumination strategies to reliably detect liquid levels and printing defects.
- **Industrial Integration:** Commissioned vision systems using smart cameras and custom illumination for auto-part recognition at a car subdivision of Renault; integrated vision sensors with robotic arms for flame-surface treatment processes.
- **QA & Feasibility:** Performed systematic Gage R&R assessments and feasibility studies for new industrial clients, ensuring 24/7 reliability of vision components on the floor.

Medical University of Innsbruck

PhD Researcher – Optical Metrology & Automation

Innsbruck, Austria

Jan 2022 – Present

- **System Build & Justage:** Led the end-to-end assembly of a custom multiphoton microscope; adjusted galvo scanners, laser sources, and PMTs to achieved sub-micron resolution.
- **Adaptive Optics Control:** Programmed a Python/C++ control stack to synchronize scientific cameras with active wavefront correctors for real-time sample characterization.
- **Vendor & Component QM:** Managed technical specification sheets and qualified precision hardware from Thorlabs and Edmund Optics as part of system procurement.

- **Instrumentation Design:** Prototyping of a dispersed fiber-probe Raman spectrometer; handled optical layout justage and hardware-software calibration.
- **Data Evaluation Hooks:** Integrated computational data analysis (Python ML) with hardware sensors to optimize measurement throughput and validation.

EDUCATION

Medical University of Innsbruck
Ph.D. in Adaptive Optics & Microscopy

Austria
Jan 2022 – Present

Friedrich Schiller University Jena
M.Sc. in Photonics

Germany
2019 – 2021

University of Cauca
B.Sc. in Physics Engineering

Colombia
2012 – 2018

PUBLICATIONS & AWARDS

- **Awards:** ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.
- Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* 12(1), 176–184 (2025).
- Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. *Applied Sciences*, 14(11), 4726 (2024).
- Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. *Biomedical Optics Express* 14(4), 1562–1578 (2023).

LANGUAGES

English (Professional),
German (B2 - Working Proficiency),
Spanish (Native)

REFERENCES

Prof. Monika Ritsch-Marte
Head of Institute of Biomedical Physics
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Medical University of Innsbruck

Dr. Christoph Krafft
Spectroscopy Group Leader
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Leibniz Institute of Photonic Technology